

The Deep Complex-Network Dichotomy Is Stable under Hausdorff-Null Modifications

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Status: candidate full result, likely valid. The complete fixed-depth dichotomy in arXiv:2012.03351 persists when a continuous activation is changed arbitrarily on a set of zero one-dimensional Hausdorff measure. This is a broad natural hypothesis strictly weaker than continuity and resolves the source's stated future-work request. It does not claim a classification of all discontinuous activations.

1 Source question and result

Felix Voigtlaender's *The universal approximation theorem for complex-valued neural networks*, arXiv:2012.03351, considers fixed-depth, arbitrarily wide complex-valued feedforward networks [1]. Write $\mathcal{NN}_{\sigma,L}^d$ for the scalar-output network class on $\mathbb{C}^d$ with activation $\sigma : \mathbb{C} \rightarrow \mathbb{C}$ and L hidden layers.

The source class $\mathcal{M}$ consists of locally bounded activations for which the closure of the ordinary discontinuity set has planar Lebesgue measure zero. Source Theorem 1.4 proves for $L \geq 2$ that $\mathcal{NN}_{\sigma,L}^d$ is universal if $\sigma \in \mathcal{M}$ satisfies neither

- (a) $\sigma(z) = p(z, \bar{z})$ almost everywhere for some $p \in \mathbb{C}[X, Y]$;
- (b) $\sigma = g$ almost everywhere or $\sigma = \bar{g}$ almost everywhere for an entire $g : \mathbb{C} \rightarrow \mathbb{C}$.

The converse assumes σ continuous. Example 4.13 shows that simply dropping continuity is impossible. The remark after Theorem 1.4, on arXiv PDF page 4, asks for natural weaker hypotheses that restore the converse.

2 Full criterion

Let $\mathcal{H}^1$ denote one-dimensional Hausdorff measure on $\mathbb{C} \cong \mathbb{R}^2$.

Theorem 1 (Hausdorff-null stability of the complete dichotomy). *Let $h : \mathbb{C} \rightarrow \mathbb{C}$ be continuous and let $\sigma \in \mathcal{M}$ satisfy*

$$\mathcal{H}^1(E) = 0, \quad E := \{z \in \mathbb{C} : \sigma(z) \neq h(z)\}.$$

For every $d \in \mathbb{N}$ and $L \geq 2$, the class $\mathcal{NN}_{\sigma,L}^d$ has the universal approximation property if and only if h is none of the following:

- (i) a function $z \mapsto p(z, \bar{z})$ with $p \in \mathbb{C}[X, Y]$;
- (ii) an entire function;

example, it is well-known (see for instance [16, Remark 2]) that bounded polyharmonic functions are necessarily constant. Thus, the theorem shows that if $\sigma \in \mathcal{M}$ is bounded but not equal to a constant (almost everywhere), then $\mathcal{NN}_\sigma^d$ is universal.

Furthermore, the theorem shows that any “bad” activation function σ necessarily has to be smooth (possibly after changing it on a null-set). Thus, any activation function that can not be made smooth by changing it on a null-set is a “universal” activation function.

Let us denote by $\mathcal{NN}_{\sigma,L}^d$ the set of complex-valued feedforward neural networks with activation function σ , input dimension d , and L hidden layers; a precise definition is given in Definition 4.9 below. The following theorem characterizes the activation functions σ for which the deep network class $\mathcal{NN}_{\sigma,L}^d$ ($L \geq 2$) is universal. Contrary to the real setting, there are strictly more “universal” activation functions for deep networks than for shallow ones.

Theorem 1.4. *Let $\sigma \in \mathcal{M}$ and assume that none of the following properties hold:*

- a) *we have $\sigma(z) = p(z, \bar{z})$ for almost all $z \in \mathbb{C}$, where $p \in \mathbb{C}[X, Y]$ is a complex polynomial of two variables,*
- b) *we have $\sigma = g$ almost everywhere or $\sigma = \bar{g}$ almost everywhere, where $g : \mathbb{C} \rightarrow \mathbb{C}$ is an entire holomorphic function.*

Then, for each $L \in \mathbb{N}_{\geq 2}$ and each $d \in \mathbb{N}$, the class $\mathcal{NN}_{\sigma,L}^d$ of deep complex-valued neural networks with activation function σ and L hidden layers has the universal approximation property.

Conversely, if $\sigma : \mathbb{C} \rightarrow \mathbb{C}$ is continuous and satisfies a) or b), then $\mathcal{NN}_{\sigma,L}^d$ does not satisfy the universal approximation property for any $d, L \in \mathbb{N}$.

Remark. Note that the necessary condition requires σ to be continuous, whereas for the sufficient condition it is only assumed that $\sigma \in \mathcal{M}$. The additional continuity assumption is not simply a proof artifact. In fact, we will see in Example 4.13 that there does exist a discontinuous activation function $\sigma \in \mathcal{M}$ that coincides almost everywhere with a polynomial $p(z, \bar{z})$, but such that $\mathcal{NN}_{\sigma,L}^d$ is nevertheless universal for all $L \in \mathbb{N}_{\geq 2}$ (but not for $L = 1$). We leave it as future work to determine natural conditions on σ that are weaker than continuity, but under which a necessary condition as in the above theorem still holds.

We would like to emphasize that any *discontinuous* function σ satisfying (at least) one of the two conditions in Theorem 1.4 has to be quite pathological: Such a function is discontinuous, but coincides almost everywhere with a continuous function. In that case, it is usually much more natural to work with the “continuous version” of σ instead of with σ itself. Moreover, such a function cannot be evaluated numerically in a reliable way and is therefore useless for practical applications.

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Figure 1: Source Theorem 1.4 and its future-work request, arXiv PDF page 4.

(iii) *the conjugate of an entire function.*

In each excluded case, the stronger source conclusion holds: there exist $F \in C_c(\mathbb{C}^d; \mathbb{R})$ and $\varepsilon > 0$, depending only on the excluded case, d , and L , such that

$$\|F - \Phi\|_{L^1(B_1(0))} \geq \varepsilon \quad \text{for every } \Phi \in \mathcal{NN}_{\sigma,L}^d.$$

The continuous representative h is unique under the hypothesis: two such representatives agree almost everywhere and hence, by continuity, everywhere.

Idea of the proof

In an obstructed case, h is real analytic. Every nonconstant intermediate preactivation is therefore real analytic as a function of the input. Such a map cannot send a positive-volume set into an $\mathcal{H}^1$ -null subset of the plane. Hence the exceptional values of σ are invisible almost everywhere along every nonconstant neuron. A constant preactivation only produces a constant, even if its value lies in E .

Replacing each nonconstant use of σ by h , while retaining the actual constant values, produces a surrogate equal to the original network almost everywhere. For polynomial h the surrogate

has a width-independent degree bound; for entire or conjugate-entire h it is holomorphic or antiholomorphic. The source's fixed-order polyharmonic obstruction applies uniformly. The converse direction already follows from the source's sufficient theorem because an $\mathcal{H}^1$ -null set is planar-null.

3 Two lemmas

Lemma 2 (pullback of a Hausdorff-null planar set). *Let $U \subset \mathbb{R}^m$ be connected and open, let $A : U \rightarrow \mathbb{R}^2$ be a nonconstant real-analytic map, and let $E \subset \mathbb{R}^2$ satisfy $\mathcal{H}^1(E) = 0$. Then $A^{-1}(E)$ has m -dimensional Lebesgue measure zero.*

Proof. Let $C = \{x \in U : DA(x) = 0\}$. Since A is nonconstant, some scalar component A_j is nonconstant, and some partial derivative of A_j is a nonzero real-analytic function. The zero set of a nonzero real-analytic function has Lebesgue measure zero [2]; hence C is null.

If $x \in U \setminus C$, then for some $j \in \{1, 2\}$ we have $\nabla A_j(x) \neq 0$. The inverse function theorem gives a neighborhood V_x on which A_j can be used as the first coordinate of a C^1 diffeomorphism. Now

$$A^{-1}(E) \cap V_x \subset A_j^{-1}(\pi_j(E)) \cap V_x,$$

where $\pi_j : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a coordinate projection. Since π_j is Lipschitz, $\mathcal{H}^1(\pi_j(E)) = 0$, and therefore $\pi_j(E)$ has one-dimensional Lebesgue measure zero. In the local coordinates above, its inverse image is contained in $\pi_j(E) \times \mathbb{R}^{m-1}$ and is null by Fubini. A countable subcover of $U \setminus C$, together with the nullity of C , proves the claim. $\square$

Lemma 3 (analytic surrogate for every obstructed network). *Let h be one of the three excluded activations in Theorem 1, let $\sigma : \mathbb{C} \rightarrow \mathbb{C}$ be arbitrary, and suppose $\mathcal{H}^1(\{\sigma \neq h\}) = 0$. For every $\Phi \in \mathcal{NN}_{\sigma, L}^d$ there is a function $Q : \mathbb{C}^d \rightarrow \mathbb{C}$ such that $\Phi = Q$ almost everywhere and:*

- (a) *if $h(z) = p(z, \bar{z})$ and $D = \max\{1, \deg p\}$, then Q is a polynomial in $z_1, \bar{z}_1, \dots, z_d, \bar{z}_d$ of degree at most D^L ;*
- (b) *if h is entire or conjugate-entire, then Q is holomorphic or antiholomorphic on $\mathbb{C}^d$.*

Proof. Fix network weights and use the source recursion

$$z^{(-1)} = z, \quad z^{(j)} = \sigma(b_j + A_j z^{(j-1)}) \quad (0 \leq j < L), \quad \Phi = b_L + A_L z^{(L-1)}.$$

We construct surrogate vectors $\tilde{z}^{(j)}$ inductively. Put $\tilde{z}^{(-1)} = z$. Given $\tilde{z}^{(j-1)}$, set

$$V^{(j)} = b_j + A_j \tilde{z}^{(j-1)}.$$

For every component $V_k^{(j)}$, define

$$\tilde{z}_k^{(j)} = \begin{cases} h(V_k^{(j)}), & V_k^{(j)} \text{ is nonconstant,} \\ \sigma(c), & V_k^{(j)} \equiv c. \end{cases}$$

We prove simultaneously that $z^{(j)} = \tilde{z}^{(j)}$ almost everywhere and that every surrogate component has the asserted analytic form. The induction hypothesis makes the actual preactivation equal to $V^{(j)}$ almost everywhere. If $V_k^{(j)}$ is nonconstant, it is real analytic, so Lemma 2 shows that $(V_k^{(j)})^{-1}(E)$ is null. Off this set and the preceding null set, $\sigma(V_k^{(j)}) = h(V_k^{(j)})$. If $V_k^{(j)} \equiv c$, the actual activation equals the constant $\sigma(c)$ almost everywhere. There are only finitely many neurons, so the vector equality follows.

If $h = p(z, \bar{z})$, this recursion consists of polynomial composition, affine combination, and constants. Its degree is multiplied by at most D at each hidden layer, giving degree at most D^L . If h is entire, all nonconstant components remain holomorphic. If h is conjugate-entire, their common orientation alternates between holomorphic and antiholomorphic at each hidden layer; constants have either orientation. Affine combinations preserve the common orientation. Thus $Q := b_L + A_L \tilde{z}^{(L-1)}$ has the claimed form and equals Φ almost everywhere. $\square$

4 Proof of the full criterion

Proof of Theorem 1. Assume first that h is excluded. Apply Lemma 3 to every $\Phi \in \mathcal{NN}_{\sigma, L}^d$. In the polynomial case, every surrogate Q has degree at most D^L , so every one-complex-dimensional slice of Q is polyharmonic of the common order $D^L + 1$. In the entire and conjugate-entire cases every such slice is harmonic. Voigtlaender’s fixed-order polyharmonic obstruction, Lemma 4.8, supplies $F \in C_c(\mathbb{C}^d; \mathbb{R})$ and $\varepsilon > 0$ with

$$\|F - Q\|_{L^1(B_1(0))} \geq \varepsilon$$

for every surrogate [1]. Since $\Phi = Q$ almost everywhere, the same lower bound holds for Φ .

Conversely, suppose h is not excluded. Since $\mathcal{H}^1(E) = 0$, we have $\sigma = h$ almost everywhere. If σ satisfied source obstruction (a) or (b), continuity would upgrade the corresponding almost-everywhere identity for h to a pointwise identity, contradicting the assumption on h . Thus neither obstruction holds for $\sigma \in \mathcal{M}$, and source Theorem 4.11 gives universality for every $d \in \mathbb{N}$ and $L \geq 2$. $\square$

5 Strictness and the source counterexample

Proposition 4. *Theorem 1 includes discontinuous obstructed activations that are not approximately continuous everywhere.*

Proof. Let $h(z) = \operatorname{Re} z = (z + \bar{z})/2$ and define

$$\sigma(z) = \begin{cases} 1, & z = 0, \\ h(z), & z \neq 0. \end{cases}$$

Then $E = \{0\}$ has $\mathcal{H}^1(E) = 0$. The activation σ is bounded on compact sets and has the single discontinuity 0, so $\sigma \in \mathcal{M}$. It is neither continuous nor approximately continuous at 0, because its approximate limit there is $0 \neq \sigma(0)$. Nevertheless, Theorem 1 gives the full non-universality conclusion for every fixed depth. This example is not covered by the earlier approximate-continuity packet. $\square$

In source Example 4.13, the activation equals $\operatorname{Re} z$ off the real axis and equals complex ReLU on the real axis. Its actual deviation set from $\operatorname{Re} z$ is the negative half-line, which has positive $\mathcal{H}^1$ measure. The second layer can map into and read that entire curve, producing universality. Thus the source counterexample lies exactly beyond the dimensional smallness imposed here.

6 Verification, scope, and novelty

The proof is deterministic and uses no computation. The main review points are: (1) the local-coordinate proof of Lemma 2; (2) the constant-component branch in Lemma 3; (3) the common degree bound D^L independent of widths and exceptional values; and (4) applying the source L^1

obstruction to the surrogates, which is legitimate because each original network agrees with its surrogate almost everywhere.

The theorem gives an exact universal/non-universal classification for all activations in $\mathcal{M}$ that are $\mathcal{H}^1$ -null modifications of a continuous activation. This fully supplies the natural weaker-than-continuity condition requested by the source. It does not classify modifications supported on sets of positive $\mathcal{H}^1$ measure or every discontinuous activation in $\mathcal{M}$.

A bounded novelty search through 9 August 2026 checked the run indexes; the exact source wording; arXiv/web queries for CVNNs with Hausdorff-null, countable, or null-set modifications; and the nearby later papers arXiv:2303.16813 and arXiv:2305.16910 [3, 4]. Those later works use smooth or continuous activations. No direct statement of the Hausdorff-null stability theorem was found. Mathematical confidence is high; novelty confidence is moderate-high.

References

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